Year 9 Mathematics

Unit 1: Number

1.1 STEM Powers of 10

STEM Powers of 10

STARTER · 5 MIN

Date: _____ **Title: STEM Powers of 10**

LEARNING OBJECTIVES

- Use and understand powers of 10
- Use the prefixes for powers of 10
- Multiply and divide by any power of 10

STARTER *Answer in your book. No calculators.*

1. Work out 68 × 100
2. Work out 62 ÷ 100
3. Work out 0.7 × 1000
4. Work out 4.97 ÷ 100

Starter — Answers

STARTER

Tick or correct in green pen

1. **6800**
2. **0.62**
3. **700**
4. **0.0497**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use and understand powers of 10	☐
SILVER	I can use the prefixes for powers of 10	☐
GOLD	I can multiply and divide by any power of 10	☐

Tick each box in the plenary when you can do it on your own.

Introduction — STEM Powers of 10

INTRODUCTION · 3 MIN

A power of 10 tells you how many 10s to multiply or divide by.

Positive powers

Multiply 10 by itself.

$10^3 = 10 \times 10 \times 10 = 1000$

Zero power

Any number to the power 0 is 1.

$10^0 = 1$

Negative powers

Divide 1 by the positive power.

$$10^{-2} = \frac{1}{100} = 0.01$$

Prefixes

kilo 10^3 mega 10^6 giga 10^9

milli 10^{-3} micro 10^{-6} nano 10^{-9}

Examples 1 and 2 — Powers of 10 as Numbers

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint $10^{-n} = \frac{1}{10^n}$ 10^n = 1 followed by n zeros

Example 1

Write 10^5 as an ordinary number.

Example 2

Write 10^{-3} as a fraction and as a decimal.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Five 10s multiplied together

$10 \times 10 \times 10 \times 10 \times 10$

= 100 000

Tip: 10^5 is 1 followed by 5 zeros.

Example 2

Negative power: divide

$$10^{-3} = \frac{1}{10^3} = \frac{1}{1000}$$

= 0.001

Tip: 10^{-3} has 3 decimal places.

Task 1

PRACTICE · 9 MIN

BRONZE

Write each as an ordinary number or a decimal

Hint $10^{-n} = \frac{1}{10^n}$ 10^n = 1 followed by n zeros

1. 10^4
2. 10^7
3. 10^2
4. 10^{-1}
5. 10^{-4}

★ 10^{-6}

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **10 000**
2. **10 000 000**
3. **100**
4. **0.1**
5. **0.0001**

★ **0.000 001**

Examples 3 and 4 — Prefixes and Units

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint

kilo 10^{3} mega 10^{6} giga 10^{9}

milli 10^{-3} micro 10^{-6} nano 10^{-9}

Example 3

Convert 2.4 MJ to joules (J).

Example 4

Convert 750 mg to grams (g).

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

mega = 10^6 = 1 000 000

MJ is bigger than J: multiply

2.4 × 1 000 000

= 2 400 000 J

Tip: bigger unit to smaller unit: multiply.

Example 4

milli = 10^{-3}: 1000 mg = 1 g

mg is smaller than g: divide

750 ÷ 1000

= 0.75 g

Tip: smaller unit to bigger unit: divide.

Task 2

PRACTICE · 9 MIN

SILVER

Convert each measurement

Hint kilo 10^{3} mega 10^{6} giga 10^{9} milli 10^{-3} micro 10^{-6} nano 10^{-9}

1. 4 kg to g
2. 12.5 GW to W
3. 3.6 km to m
4. 450 mg to g
5. 2 500 000 µm to m

★ 8 000 000 000 nm to m

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **4000 g**
2. **12 500 000 000 W**
3. **3600 m**
4. **0.45 g**
5. **2.5 m**

★ **8 m**

Examples 5 and 6 — Comparing Units

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint kilo 10^3 mega 10^6 giga 10^9 milli 10^{-3} micro 10^{-6} nano 10^{-9}

Example 5 (2 marks)

How many times bigger is a kilometre than a millimetre?

Example 6 (2 marks)

Write 0.42 mm in micrometres (μm).

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Subtract the powers of 10

km = 10^{3} m, mm = 10^{-3} m

$10^{3} \div 10^{-3} = 10^{6}$

1 000 000 times

Tip: 3 − (−3) = 6

Example 6

mm = 10^{-3} m, µm = 10^{-6} m

$10^{-3} \div 10^{-6} = 10^{3}$

0.42 × 1000

= 420 µm

Tip: smaller unit, so the number gets bigger.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint
kilo 10^{3} mega 10^{6} giga 10^{9}
milli 10^{-3} micro 10^{-6} nano 10^{-9}

1. How many mm in 1 km?

2. How many W in 1 GW?

3. How many times bigger is a megawatt than a kilowatt?

4. Write 0.42 mm in μm

5. Write 3.5 km in mm

★ Write 50 000 nm in mm

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. **1 000 000**
2. **1 000 000 000**
3. **1000 times**
4. **420 µm**
5. **3 500 000 mm**

★ **0.05 mm**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Write 10^{-2} as a decimal.

SILVER Convert 3.5 km to metres.

GOLD How many times bigger is a megawatt than a kilowatt?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **0.01**

SILVER **3500 m**

GOLD **1000 times**

Exam Practice — Powers of 10

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $10^{-n} = \frac{1}{10^n}$ 10^n = 1 followed by n zeros

Question 1

(2 marks)

Write as an ordinary number.

(a) 10^6 (1)

.....

(b) 10^{-3} (1)

.....

Question 2

(2 marks)

Write 10^{-5}

(a) as a fraction (1)

.....

(b) as a decimal (1)

.....

Exam Practice — Prefixes and Units

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint kilo 10^{3} mega 10^{6} giga 10^{9} milli 10^{-3} micro 10^{-6} nano 10^{-9}

Question 3

(2 marks)

A tank holds 4.5 kilolitres of water.

How many litres is this?

..... litres

Question 4

(2 marks)

A virus is 2500 nm long.

Write its length in metres.

..... m

Exam Practice — Comparing Units

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint kilo 10^{3} mega 10^{6} giga 10^{9} milli 10^{-3} micro 10^{-6} nano 10^{-9}

Question 5 (2 marks)

How many times bigger is a gigawatt than a kilowatt?

.....

Question 6 (2 marks)

A virus is 0.000 12 mm wide.
Write its width in nanometres (nm).

..... nm

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 1 000 000 — B1

(b) 0.001 — B1

Tip: $10^{-3} = 1 \div 1000$

Question 2

(a) $\dfrac{1}{100\,000}$ — B1

(b) 0.000 01 — B1

Tip: 10^{-5} has 5 decimal places.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

kilo = 10^3 = 1000 — M1

4.5 × 1000 = 4500 litres — A1

Tip: bigger unit to smaller unit: multiply.

Question 4

nano = 10^{-9}: ÷ 1 000 000 000 — M1

= 0.000 002 5 m — A1

Tip: smaller unit to bigger unit: divide.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$10^9 \div 10^3$ — M1

$= 10^6 = 1\ 000\ 000$ times — A1

Tip: giga = 10^9, kilo = 10^3

Question 6

$1\ \text{mm} = 10^6\ \text{nm}$ — M1

$0.000\ 12 \times 1\ 000\ 000 = 120\ \text{nm}$ — A1

Tip: smaller unit, so the number gets bigger.